Jinhong Du

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Target Position: Embodied Intelligence / Robotic Manipulation Algorithm Engineer

Focus: Autonomous manipulation in complex, densely cluttered scenes, combining 3D geometric perception, push-grasp coordination, and reinforcement learning for decision-making. End-to-end experience in algorithm design, simulation training, and real-robot deployment.

Education

South China University of Technology (SCUT) Sep. 2024 – Jun. 2027

School of Automation Science and Engineering

M.Eng. in Electronic Information, Top 10%

Jinan University Sep. 2020 – Jun. 2024

International Energy College

B.Eng. in Automation, Top 7%

Publications

- **GAPG: Geometry Aware Push-Grasping Synergy for Goal-Oriented Manipulation in Clutter**.
  First Author; **IEEE ICRA 2026**, Accepted. [arXiv] [GitHub] [Demo]
- **SuperGrasp: Single-View Object Grasping via Superquadric Similarity Matching, Evaluation, and Refinement**.
  First Author; **IEEE RA-L**, Accepted. [arXiv]
- **PushPoint: Point-Conditioned Action Learning for Graspability-Aware Bin Rearrangement**.
  First Author; **Under Review**.

Project Experience

GAPG – Goal-Oriented Push-Grasping in Clutter | Lead Jun. 2025 – Sep. 2025

PointNet++ / PyBullet / ROKAE / ICRA 2026 [Open-source code]

National Key R&D Program of China | Autonomous Intelligence and Emergent Swarm Intelligence in Multi-Robot Collaborative Manufacturing

- Developed a geometry-aware push-grasp framework to recover from single-step grasp failures on occluded or stacked targets, jointly evaluating grasp feasibility and pushing utility.
- Sampled 6D grasp candidates with GraspNet; evaluated gripper and closing-region point clouds using PointNet++; represented push poses as points and encoded geometric relations between the target and surrounding objects.
- Trained grasping and pushing modules on 40k samples each and deployed to ROKAE xMate ER7Pro via zero-shot sim-to-real transfer, achieving **90.4% grasp success** and **95.0% task completion** on 15 unseen objects in dense real-world clutter.

SuperGrasp – Single-View Grasp Candidate Generation, Evaluation, and Refinement | Lead Oct. 2025 – Mar. 2026

Superquadric / ROS / ROKAE / RA-L

National Science and Technology Major Project: Intelligent Manufacturing Systems and Robotics
Cyber-Physical Integration for Customized Manufacturing Processes

- Addressed unstable grasping caused by incomplete single-view geometry using superquadric parameter matching to generate grasp candidates, followed by E-RNet evaluation and refinement.
- Built an offline library of 1.2k geometric primitives to transfer pre-grasp poses via parameter matching; designed anchor-guided E-RNet to fuse closing-region geometry with surrounding context and predict grasp and refinement feasibility.
- Trained E-RNet on 100k stable grasp samples and deployed to ROKAE xMate ER7Pro via zero-shot sim-to-real transfer, achieving **94.12% grasp success**, **96.00% task success**, and **0.713 s inference time** on 10 unseen objects in dense real-world scenes.

Project Experience (continued)

Highly Autonomous Picking for Massive-SKU Warehousing | Lead Apr. 2026 – Present

SAC / PyBullet / UR5e

Industry Collaboration | Leading intelligent warehousing company; million-RMB-scale collaboration

- Extended SuperGrasp to SuperGrasp++ for dense-bin picking, jointly predicting gripper-width corrections and collision risk to generate reliable grasps despite severe stacking, occlusion, and wall-adjacent targets.
- Developed PushPoint to model spatial relations among the target, neighboring objects, and bin walls; designed the state representation, hybrid discrete–continuous action space, and grasp-score reward; trained a goal-directed non-prehensile policy with SAC to restore target graspability.
- Trained grasping and non-prehensile policies and deployed them to UR5e via zero-shot sim-to-real transfer; on 50+ unseen industrial objects, improved bin-clearing success from **80%** with SuperGrasp++ alone to **95%** by integrating PushPoint.

Technical Skills

- **Technical Stack:** Python, PyTorch, ROS, MoveIt, PyBullet, Isaac Gym, Open3D, SolidWorks.
- **Other:** English CET-6; IROS 2026 Reviewer.